Towards SOFT ELECTRONICS For Shape-Shifting Circuits

Dr S.S. Verma is a professor at Department of Physics, Sant Longowal Institute of Engineering and Technology, Sangrur, Punjab

Science fiction is inching closer to reality with the development of revolutionary self-propelling liquid metals—a critical step towards future elastic electronics. At present, electronic devices like smart phones and computers are mainly based on circuits that use solidstate components, with fixed metallic tracks and semiconducting devices. This leads to rigid structures. Besides, none of the current technologies is able to create homogeneous surfaces of atomically thin semiconductors on large surface areas that are useful for industrial-scale chip fabrication.

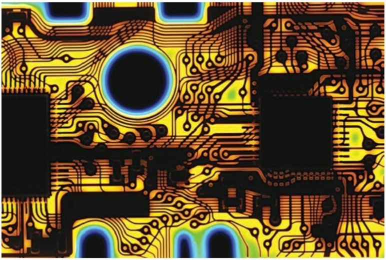
Advancements in electronics can be made through flexible and dynamically reconfigurable soft circuit systems. This involves taking electronics beyond the confines of solidstate circuits to soft-state circuits. The goal is to create elastic electronic components. These are a type of soft circuit systems that would function in a way analogous to biological cells. That means electronic circuits that can move independently and also interact with each other to construct new circuits. To achieve this, liquid metals are required.

The use of liquid metals in electronics has always fascinated the scientists due to their inherent advantages of fluidity and shape-shifting qualities. However, there are issues to be dealt with, like availability of only a few pure liquid metals at room temperature. Among liquid metals, non-toxic alloys of gallium so far seem to be the most promising candidate for realising the dream of truly elastic electronic components.

Gallium alloys: Liquid metals fit for electronics

Though all the pure metals listed in the table (next page) are liquid at room temperature, these are not suitable for use in electronics due to reactivity and stability problems associated with them. Liquid metals also consist of alloys with very low melting points, which form an eutectic that is liquid at room temperature. The standard metal of choice used to be mercury, but gallium-based alloys, which are lower both in their vapour pressure at room temperature and toxicity, are being used as a replacement in various applications.

Gallium shares similarities with aluminium, indium and thallium. It is commonly used to make alloys with low melting points, which are predominantly used in electronics. For example, gallium arsenide is used in microwave circuits. Some of the common gallium alloys are Ga-In-Sn, Ga-Al, Ga-In, Ga-Sb, Ga-Sn, Ga-Pb, Ga-Mg, Ga-Bi and Ga-Bi-Sn.



Static circuit boards may one day give way to dynamic, changeable ones that can pop up on demand (Image courtesy: https://cosmosmagazine.com)